



Effects of bedding type and forage provision on growth performance, health, and rumen fermentation in dairy calves

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ABSTRACT

The objective of this study was to determine the effects of different diets (concentrate [CON] or concentrate with oat hay [FOR]) and bedding materials (straw [STR] or rice husks [RIC]) on growth performance, rumen fermentation, and blood metabolites in dairy calves. Holstein dairy calves (initial BW 41.2 ± 4.03 kg; $n = 15$ calves per treatment) were randomly assigned to 4 dietary treatments in a 2×2 factorial design: (1) CON-STR, (2) CON-RIC, (3) FOR-STR, and (4) FOR-RIC. Calves were offered similar volumes of pasteurized whole milk twice daily until complete weaning at the end of wk 8. All calves received a commercial pelleted starter concentrate, whereas those fed forage received oat hay for 12 wk. The bedding materials were renewed weekly. From wk 13 to 28, all calves were housed together and fed identical diets. Feed intake and refusals were recorded daily for wk 1 to 12. Body weight and structural measurements were taken on the day of birth and subsequently, every 2 wk until wk 12; BW was also recorded on the last day of wk 28. Rumen samples were taken at the end of wk 1, 5, 12, and 28 of age via esophageal tube, and blood samples were taken at the end of wk 12 for determination of metabolic and immunity related indicators. Three-way interactions (diet $\times$ bedding $\times$ time) were observed for milk and solid feed nutrient intakes, rumen pH, acetate and total VFA (TVFA). The FOR group had greater milk intake, total solid feed DMI, total DMI, and ADG than the CON group from wk 1 to 12, leading to higher BW at the end of wk 28. Calves on RIC bedding consumed more forage compared with those on STR during wk 1 to 12. Although the FOR group had greater dietary NDF, physi-

cally adjusted NDF, and ADF intakes compared with CON from wk 1 to 8, the FOR-RIC calves showed the highest intakes and CON-RIC had the lowest (wk 9 to 12 and wk 1 to 12). A significant bedding $\times$ day interaction was observed for ADG, withers height, and heart girth. Compared with RIC-bedded calves, STR-bedded calves had lower ADG for wk 1 to 8 but higher ADG for wk 9 to 12. Straw bedding was associated with lower blood urea and IL-6 and higher total antioxidant capacity compared with RIC. Rumen pH was higher in STR-bedded calves than in RIC-bedded calves, with no effect of diet. Calves on CON-STR had the lowest concentration of acetate and TVFA at the end of wk 12. In conclusion, this study showed rice husk bedding had significant effect on dietary intake, calf health and immunity, and performance of the calf during wk 1 to 28. In contrast, forage feeding can mitigate the negative effects, especially during wk 9 to 12.

Key words: calf, forage feeding, bedding type

INTRODUCTION

Heifer calves are a key investment in dairy enterprises. Proper nutrition and management form the basis for raising healthy animals that are able to produce more milk during their productive lives (Gelsinger et al., 2016; Zhang et al., 2024), as well as for enhancing herd longevity (Heinrichs and Heinrichs, 2011). In the first month of life, calves depend mainly on an adequate amount of milk for maintenance and growth, as they cannot fully utilize concentrates or forage (Khan et al., 2011b; Miller-Cushon et al., 2013). Strategies for supplementing forage to optimize consumption of solid feed in dairy calves have continued to develop over the past few decades. Although some studies have previously opposed the use of forage, a growing body of evidence indicates that forage provision can improve growth performance, rumen fermentation, and calf health (Castells et al., 2015;

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

Hill et al., 2019). Meta-analytic studies have reported that different forages modulate rumen pH and have the potential to alter starter feed intake and performance in calves (Imani et al., 2017; Xiao et al., 2025b), probably as a result of improved rumination (Xiao et al., 2025a). Furthermore, forage has been reported to modulate behavior and improve welfare indicators (Hill et al., 2019; Horvath and Miller-Cushon, 2019). Some studies have sought to establish whether forage consumption can affect rumen fermentation in calves (Khan et al., 2011a; Daneshvar et al., 2015), as well as modulate behavior, health, and performance (Hill et al., 2019; Horvath and Miller-Cushon, 2019), thus improving calf welfare. However, the optimal utilization of forage depends on several factors, including forage quality, the physical form of the starter feed, forage level, feeding method, and milk level (Xiao et al., 2025b). Among these factors, the housing environment, in particular the bedding material, may interact with forage provision, but this has received limited attention.

Bedding material is a key environmental factor influencing calf comfort, health, and welfare. Calves are raised on diverse bedding materials, such as rice hulls, wheat straw, wood shavings, concrete, granite fines, and sand (Abuelo et al., 2019). Bedding type affects weight gain, health, and welfare (Hill et al., 2011; Shivley et al., 2018) in calves. Soft, dry materials such as sawdust are generally preferred over harder surfaces such as concrete (Camiloti et al., 2012). Notably, calves tend to nibble on bedding material during the preweaning period, especially when provided with plant-based materials (Diaz et al., 2001; Hill et al., 2019). Reports indicate that some calves begin consuming straw bedding within wk 1 of life (Wang et al., 2022), indicating that exploratory ingestion of bedding starts very early. This behavior is highly relevant to calf nutrition because different bedding types vary in palatability and nutritional content. Straw is readily consumed by calves and provides substantial physically adjusted NDF (**paNDF**), whereas rice husk contains minimal paNDF and is far less palatable. Excessive straw ingestion has been shown to impair nutrient intake and potentially hinder growth (Gasiorek et al., 2022). In contrast, calves bedded on rice husks are likely to ingest negligible bedding material, allowing dietary treatment effects to be assessed without confounding contributions from incidental fiber intake. This suggests that bedding edibility may interact directly with forage provision and influence calf nutritional responses.

Beyond nutritional confounding, bedding type affects hygienic conditions. Certain bedding materials can harbor pathogenic bacteria (Wolfe et al., 2018), especially when they remain wet for a long time. Long wheat straw has been shown to support higher coliform counts than rice hulls, sand, or wood shavings after prolonged use

(Panivivat et al., 2004). Despite these implications, the combined influence of forage feeding and bedding material on calf performance, health, and rumen function has not been evaluated. This is an important knowledge gap because diet and bedding are often considered independently, even though calves' bedding ingesting behavior suggests they likely interact.

Rice husk possesses several characteristics that may mitigate concerns associated with straw bedding. First, its hardness and high lignin content make it less palatable, reducing the likelihood of excessive bedding consumption. Second, the physical properties of rice husks promote excellent drainage and moisture resistance, promoting drier, cleaner resting surfaces that reduce pathogen load and disease risk (Lundborg et al., 2005; Kaufman et al., 2015). These properties make rice husks an attractive bedding material for studying diet-bedding interactions without the confounding effects of bedding ingestion.

Therefore, we hypothesized that the use of rice husks as bedding could support calf health and growth by reducing the intake of nutritionally inadequate bedding material and reducing exposure to bedding-associated pathogens. The objective of this study was to determine the interaction effect between diet type (concentrate alone vs. concentrate with hay) and bedding material (straw vs. rice husks) on growth performance, health, blood biomarkers, and rumen fermentation in dairy calves.

MATERIALS AND METHODS

Animals, Housing, and Dietary Treatments

This study was conducted at the Sichuan New Hope Huaxi Farm (Chengdu, Sichuan, China) from November 2024 to May 2025. The experiment and animal procedures were conducted according to the Regulation of the Laboratory Animals (Government of China, 2017) promulgated by Decree No. 676 of the State Council, China. The protocol used for the use and care of the animals was approved by the Animal Care and Use Committee of Sichuan Agricultural University (Chengdu, China; approval no. 20240556). The required number of animals per treatment group was estimated using a priori power analysis conducted with G*Power software (version 3.1; <http://www.gpower.hhu.de/>). Average daily gain (expressed in kg/d) was selected as the primary outcome variable to determine an appropriate sample size. Based on previous studies (Panivivat et al., 2004), the assumed SD for ADG was 0.10 kg/d, and the expected difference between groups was set at 0.20 kg/d, resulting in an effect size (Cohen's *d*) of 0.20. Using a significance level (α) of 0.05 and a desired statistical power of 0.80, the analysis indicated that a minimum of 44 calves (11 per group) was

needed. To further ensure sufficient statistical power, a total of 60 newborn Holstein calves (15 per group) were enrolled in the study. Using a randomization procedure based on date of birth, calves of comparable BW were selected daily (4, 8, or 12 calves per day) and randomly allocated to one of the 4 experimental groups ($n = 15$ per group), resulting in 4 groups balanced by birth weight (41.5 ± 4.03 kg). The experiment design was a 2×2 factorial arrangement of dietary treatments, consisting of 2 different diets (concentrate [CON] or concentrate with oat hay [FOR]) and 2 different bedding materials (straw [STR] or rice husks [RIC]). The allocation thus yielded 4 treatment groups: (1) calves fed concentrates only and raised on straw (CON-STR); (2) calves fed concentrates only and raised on rice husks (CON-RIC); (3) calves fed concentrates and forage and raised on straw (FOR-STR) and (4) calves fed concentrates and forage and raised on rice husks (FOR-RIC). Straw was used as bedding material with a depth of 20 cm. Particle size distribution, determined with a Penn State Particle Separator, was as follows: 57.2% long particles (>19 mm), 25.2% medium particles (8–19 mm), 9.8% short particles (1.18–8 mm), and 6.3% fine particles (<1.18 mm). Oat hay was cut to less than 3 cm using a stationary mixer (20 m³, Trioliet Co. Ltd., the Netherlands). The chopped oat hay had the following particle size distribution: 37.5% long particles (>19 mm), 28.3% medium particles (8–19 mm), 19.4% short particles (1.18–8 mm), and 14.8% fine particles (<1.18 mm).

Immediately after birth, calves were separated from their dams and transferred to individual hutches. They were fed 4 L of high-quality colostrum ($>22\%$ Brix) within 1 h of life. Successful passive transfer of immunity at 24 h after feeding was checked in serum total protein (TP; ≥ 5.5 g/dL). Calves were fed pasteurized whole milk twice daily at 0800 and 1500 h from wk 1 to 8. Calves were provided 8 L of milk daily from wk 1 to wk 3. This was increased to 10 L/d for wk 4 through 6, then decreased to 8 L/d in wk 7. A step-down weaning protocol was initiated at the beginning of wk 8. Milk allowance was reduced by 2 L every 3 d until the calves were fully weaned (6 L/d on d 50–52, 4 L/d on d 53–55, and 2 L/d on d 56). All calves were offered a complete pellet starter concentrate (Sichuan New Hope Huaxi Farm, China). Half of the calves were offered forage (oat hay) in addition to concentrate until wk 12. Calves fed concentrates only had access to one bucket, and those fed concentrate and hay had 2 buckets hung inside the hutch, and 2 additional buckets for water and milk were attached on the outside and in front of the hutch. Calves were raised in sterilized hutches (1.15 m wide $\times$ 1.8 m long) with an attached outdoor fenced area (~ 1.6 m $\times$ 1.5 m) from wk 1 to 12. The hutches were bedded with either

Table 1. Chemical composition of the experimental starter, hay, milk, and TMR

Item	Calf starter ¹	Oat hay ²	Milk	TMR ³
Chemical composition				
Dry matter (%)	89.0	88.1	13.2	51.2
Crude protein (%DM)	21.1	6.4	25.9	17.9
Fat (% DM)	—	—	29.5	—
SNF (% DM)	—	—	70.5	—
Starch (%DM)	30.8	0.7	—	18.0
NDF (%DM)	18.4	46.7	—	33.5
ADF (%DM)	9.7	25.7	—	25.0
ASH (%DM)	6.6	4.8	—	—
EE (%DM)	2.2	1.5	—	2.9

¹Calf starter contained 35.0% corn, 29.5% soybean meal, 13.0% soybean hulls, 10.0% wheat bran, 8.0% wheat flour, 0.5% calcium carbonate, and 4% premix compound (contained vitamin A 250,000 IU/kg, vitamin D 40,000 IU/kg, vitamin E 2,000 IU/kg, manganese 0.6 g/kg, iron 0.4 g/kg, copper 0.5 g/kg, zinc 2 g/kg, cobalt 4 mg/kg, iodine 16 mg/kg, and selenium 4 mg/kg) on a dry matter basis.

²Oat hay was cut to less than 3 cm using a stationary mixer (20 m³, Trioliet Co. Ltd., the Netherlands).

³The TMR was composed of 37.0% alfalfa hay, 32.0% corn silage, 9.5% soybean meal, 5% rapeseed meal, 7.5% steam-flaked corn, 6% corn, and 3.0% premix compound (contained vitamin A 300,000 IU/kg, vitamin D 80,000 IU/kg, vitamin E 2,000 IU/kg, Mn 2.0 g/kg, Cu 0.7 g/kg, Zn 2.5 g/kg, Co 20 mg/kg, iodine 15 mg/kg, and selenium 14 mg/kg) on a DM basis.

straw or rice husks, with the bedding renewed weekly. Beginning at wk 13, all calves were transferred to a barn and housed in one group and fed the same diet: a top-dressed ration (the same concentrate spread over hay in a ratio of 9:1; wk 13–19) and a novel TMR (wk 20–28). Feeds and water were available ad libitum. The dietary compositions are reported in Table 1. In this experiment, the detailed health status of the calves was recorded and is available in the Figshare data repository (<https://doi.org/10.6084/m9.figshare.31154191>).

Feed Sampling and Analysis

Milk, concentrate starter, and oat hay intakes were recorded as differences between the amounts offered and refused by each calf from wk 1 to 12. Sufficient amounts of concentrate starter and hay were given to allow for at least 10% refusals. The ratio of forage to total solid feed intake (FIR) was calculated based on hay intake divided by total solid feed intake (including concentrate and hay, DM basis) from wk 1 to 12. Once a week, milk and feed samples were collected and immediately frozen at -20°C . The samples were later analyzed for DM (method 930.15), CP (method 988.05), ether extract (EE; method 920.39) and crude ash (method 942.05) following AOAC International (2005) methods for concentrate starter and oat hay samples. Furthermore, starch content was analyzed using total starch assay kit (Megazyme, Bray, Ireland; AOAC method 996.11), the NDF and ADF

were analyzed using an ANKOM A2000i fiber analyzer (American ANKOM, Macedon, NY), as described by Van Soest et al. (1991).

Body Weight and Structural Measurements

Body weight was recorded on a digital scale every 2 wk until wk 12 and once at the end of wk 28 to evaluate the long-term effects of the preweaning and postweaning treatments. At the same time structural measurements (withers height and heart girth) were recorded, excluding the end of wk 28. Average daily gain was calculated as the difference in BW between 2 consecutive days in which the measurements were taken. Likewise, feed efficiency (FE; kg DMI/kg BW gain) per calf was calculated.

Rumen Fluid and Blood Sample Collection

Rumen fluid was collected via flexible esophageal tube (2-mm of wall thickness and 6-mm i.d.; SciTech Co. Ltd., Wuhan, Hubei, China) and filtered through 4 layers of cheesecloth at the end of wk 1, 5, 12, and 28 of age, 2 h after the morning feeding. The first 10 mL of rumen fluid was discarded to avoid saliva contamination. The pH value was immediately determined with a glass electrode pH meter (HORIBA Advanced Techno Co. Ltd., Osaka, Japan). The samples (15 mL) were stored at -20°C for later analysis of TVFA concentration using gas chromatography with 2-ethylbutyric acid as an internal standard.

Blood samples were collected into serum evacuated tubes containing no anticoagulant from the jugular vein at the end of wk 12 before the morning feeding. The blood was then centrifuged at $3,500 \times g$ for 15 min at 4°C and then partitioned into aliquots and stored in 1.5 mL microcentrifuge tubes at -20°C for later analysis. A comprehensive panel of biochemical parameters was measured to assess metabolic status, inflammatory response, and oxidative stress. Metabolic indicators included energy metabolites (glucose, nonesterified fatty acids [NEFA], BHB), protein metabolism and nutritional markers (urea, TP, albumin [ALB]), lipid profiles (total cholesterol [TC]), and liver function (total bilirubin [TBill]). In addition, systemic inflammation was evaluated via acute-phase proteins (APP; serum amyloid A [SAA], haptoglobin), inflammatory cytokines (TNF- α , IL-6, IL-10), and endotoxin-related markers (LPS, lipopolysaccharide-binding protein [LBP]). Oxidative stress status was determined by measuring antioxidant enzyme activities (glutathione peroxidase [GSH-Px], total superoxide dismutase [T-SOD]), lipid peroxidation product (malondialdehyde [MDA]), and total antioxidant capacity (T-AOC). The blood urea, TC, TBill, TP, and ALB concentrations were measured by an automatic biochemical analyzer

(Hitachi 7600, Hitachi High-Technologies Corporation, Tokyo, Japan) in accordance with the manufacturer's protocols. Oxidative status biomarkers (MDA, T-SOD, GSH-Px, and T-AOC) were analyzed by thiobarbituric acid, hydroxylamine, calorimetry, and 2,2'-azino-bis(3-ethylbenzothiazoline-6-sulfonic acid) (ABTS) methods using assay kits (A003-1-2, A001-1-2, A005-1-2, and A015-2-1, respectively; Jiancheng Bioengineering Institute, Nanjing, China). The concentrations of APP, such as SAA, haptoglobin and LBP, were analyzed using ELISA kits H134, H136, and H253 (Jiancheng Bioengineering Institute, Nanjing, China) in an enzyme-immunoassay instrument (Multiskan MK3, ThermoFisher Scientific Co. Ltd., Franklin, MA). The concentrations of IL-6, IL-10, and TNF- α in serum were measured using ELISA kits KJEIA0006D, KJEIA0008D, and KJEIA0018D, respectively (Beijing Kangjia Hongyuan Biotechnology Co. Ltd., Beijing, China) according to the manufacturer's instructions. Concentrations of LPS were determined by a chromogenic kinetic Limulus amoebocyte lysate assay (EC80545S, Xiamen Bioendo Technology Co. Ltd., Xiamen, China) in accordance with the manufacturer's protocols.

Statistical Analysis

Data for feed (milk, concentrate starter, hay) intake were summarized weekly. Before analyses, all data were screened for normality using the UNIVARIATE procedure of SAS. The data were analyzed using MIXED procedure of SAS (SAS 9.4, SAS Institute Inc., Cary, NC) over different stages (wk 1–8, wk 9–12, wk 1–12, and wk 1–28). Dependent variables (milk, concentrate starter and hay intakes, nutrient intake, BW, ADG, FIR, structural measurements, FE, and fermentation parameters) were treated as repeated measures per period. The linear mixed model included the fixed effects of diet, bedding material, time, and their interactions (diet $\times$ bedding, diet $\times$ time, bedding $\times$ time, diet $\times$ bedding $\times$ time). Calf was included as a random effect. Furthermore, data for wk 1 to 12 (except for BW, ADG, and rumen fermentation) and wk 1 to 28 (BW, ADG and rumen fermentation) were tested using the same model but with time and 2- and 3-way interactions effects included. Three variance-covariance structures (simple, autoregressive type 1, and compound symmetry) were tested, and the covariance structure with the lowest Akaike and information criterion was used. The calf was the subject of the repeated statement. For the blood parameters, sampled only at the end of wk 12, the model included the fixed effects of diets, bedding materials, and the interaction between diets and bedding materials. The least squares means were separated by the PDIF statement, and the Tukey–Kramer adjustment was applied to account for the multiple comparisons.

Table 2. Effect of diet and bedding material on solid and liquid DMI¹

Item ³	CON		FOR		SEM	P-value ²					
	STR	RIC	STR	RIC		DIET	BED	DIET × BED	DIET × T	BED × T	DIET × BED × T
Milk DMI (g/d)	1,138.3	1,137.7	1,155.7	1,168.9	11.6	0.04	0.59	0.55	0.42	0.14	0.02
Milk intake (kg/d)	8.6	8.6	8.8	8.9	0.1	0.04	0.58	0.55	0.14	<0.01	0.02
Starter DMI (g/d)											
wk 1 to 8	194.5	184.5	138.9	160.6	23.7	0.10	0.81	0.51	0.04	0.02	0.97
wk 9 to 12	2,405.0	2,136.4	2,274.6	2,357.7	106.9	0.67	0.39	0.11	0.02	<0.01	0.47
wk 1 to 12	931.3	835.1	850.8	893.0	48.3	0.82	0.58	0.16	0.09	<0.01	0.02
Hay DMI (g/d)											
wk 1 to 8	0.0	0.0	29.2	47.6	4.93	<0.01	0.07	0.07	<0.01	0.02	0.02
wk 9 to 12	0.0	0.0	272.5	386.4	27.8	<0.01	0.05	0.05	<0.01	0.51	0.51
wk 1 to 12	0.0	0.0	110.3	160.5	11.5	<0.01	0.04	0.04	<0.01	<0.01	<0.01
Total solid feed DMI (g/d)											
wk 1 to 8	194.5	184.5	168.1	208.2	25.0	0.96	0.55	0.32	0.94	<0.01	0.84
wk 9 to 12	2,405.0 ^{ab}	2,136.4 ^b	2,547.0 ^{ac}	2,744.1 ^c	113.0	<0.01	0.75	0.04	<0.01	<0.01	0.28
wk 1 to 12	931.3	835.1	961.1	1,053.5	51.3	0.02	0.97	0.07	<0.01	0.02	<0.01
Total DMI (g/d)											
wk 1 to 8	1,332.8	1,322.2	1,323.9	1,377.1	28.3	0.42	0.46	0.26	0.93	0.02	0.22
wk 9 to 12	2,405.0 ^{ab}	2,136.4 ^b	2,547.0 ^{ac}	2,744.1 ^c	113.0	<0.01	0.75	0.04	<0.01	<0.01	0.28
wk 1 to 12	1,690.1	1,593.6	1,731.6	1,832.8	53.2	0.01	0.97	0.07	<0.01	0.02	<0.01
FIR (%)											
wk 1 to 8	0	0	0.20	0.25	0.02	<0.01	0.27	0.27	0.02	0.95	0.95
wk 9 to 12	0	0	0.11	0.14	0.01	<0.01	0.12	0.12	0.20	0.64	0.64
wk 1 to 12	—	—	0.17	0.21	0.01	<0.01	0.17	0.17	<0.01	0.99	0.99

^{a-c}Within a row, means with different superscripts differ significantly ($P < 0.05$). Letter differences arise from a significant DIET × BED interaction.

¹CON = calves fed only starter concentrate; FOR = calves fed starter concentrate with oat hay; STR = straw bedding; RIC = rice husk bedding.

²DIET = dietary treatment (CON, FOR); BED = bedding treatment (STR, RIC); T = time. For all variables, data were summarized weekly; data were analyzed for wk 1 to 8, wk 9 to 12, and wk 1 to 12.

³FIR = forage/total solid feed intake ratio; total solid feed DMI = starter DMI + hay DMI; total DMI = milk DMI + total solid DMI.

Significance was declared at $P < 0.05$ and a trend was declared at $0.05 \leq P < 0.10$. The individual calf was the experimental unit.

RESULTS

Dietary Intake

A 3-way interaction was observed for starter DM, hay DM, total solid feed DM, and total DM intakes from wk 1 to 12, as well as milk and hay DMI during wk 1 to 8 ($P \leq 0.04$; Table 2 and Figure 1). The FOR group had greater milk intake, irrespective of their bedding material, compared with the CON group ($P = 0.04$). Similarly, total solid feed DMI ($P = 0.02$) and total DMI ($P = 0.01$) were greater for the FOR group compared with the CON group for wk 1 to 12, which could be ascribed to improved intakes from wk 9 to 12. During wk 1 to 8, the total solid feed DMI was relatively low (ranging from 184.5 to 208.2 g/d). Starting from wk 9, the total solid feed DMI increased significantly (reaching 2,136–2,744 g/d in wk 9–12; $P < 0.01$). The RIC group consumed more forage (~50 g/d) compared with the STR group ($P = 0.04$) from wk 1 to 12. Compared with the FOR group, the CON group tended to have greater starter DMI

from wk 1 to 8 ($P = 0.10$). During wk 1 to 12, a diet × time interaction was observed for FIR ($P < 0.01$).

A 3-way interaction was observed for calf period CP, EE, starch, NDF, paNDF, and ADF intakes ($P \leq 0.02$; Table 3). During wk 1 to 12, the FOR-RIC calves consumed the highest amount of dietary NDF, paNDF and ADF, whereas CON-RIC had the least ($P \leq 0.03$). These differences were more evident from wk 9 to 12 ($P = 0.02$). The FOR calves consumed more NDF, paNDF, and ADF than those on CON from wk 1 to 8 ($P \leq 0.04$). The FOR group consumed more EE compared with CON group from wk 9 to 12, and wk 1 to 12 ($P \leq 0.05$). However, CP and starch were neither affected by diet nor bedding material in any of the periods.

Growth Parameters

Table 4 shows that the initial BW was not different between the groups (41.2 ± 4.03 kg; $P = 0.35$). No significant 3-way interactions were observed for overall BW and ADG from wk 1 to 4. Similarly, for the period from wk 1 to 12, no significant 3-way interactions were found for BW, ADG, withers height, heart girth, or FE. During wk 1 to 8 and wk 9 to 12, no significant differences in the 3-way interactions were observed, except for ADG from

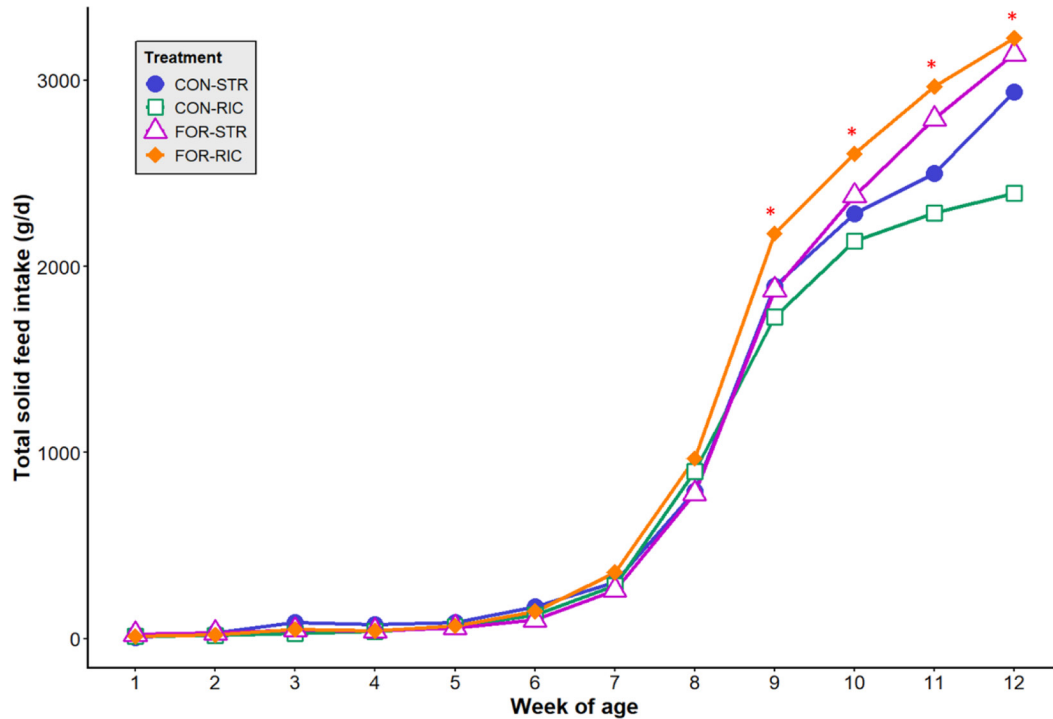


Figure 1. Mean weekly total solid feed DMI (g/d). CON-RIC = calves were fed a calf starter only and bedded on rice husks; CON-STR = calves were fed a calf starter and bedded on straw; FOR-RIC = calves were fed forage in addition to a calf starter and bedded on rice husks; FOR-STR = calves were fed forage in addition to a calf starter and bedded on straw. Asterisks (*) indicate significant differences ($P < 0.05$). Significant group comparisons: Week 9: CON-STR and FOR-RIC: $P = 0.04$; CON-RIC and FOR-RIC: $P < 0.01$. Week 10: CON-STR and FOR-RIC: $P = 0.02$; CON-RIC and FOR-RIC: $P < 0.01$. Week 11: CON-STR and FOR-STR: $P = 0.03$; CON-STR and FOR-RIC: $P < 0.01$; CON-RIC and FOR-STR: $P < 0.01$; CON-RIC and FOR-RIC: $P < 0.01$. Week 12: CON-STR and CON-RIC: $P < 0.01$; CON-STR and FOR-RIC: $P = 0.04$; CON-RIC and FOR-STR: $P < 0.01$; CON-RIC and FOR-RIC: $P < 0.01$.

wk 1 to 8 ($P = 0.04$). However, a significant diet $\times$ day interaction was observed for BW only ($P = 0.01$). At the end of 28 wk, heifers fed forage were heavier than CON calves by 10.7 kg (219.0 vs. 229.7 kg; $P = 0.01$). The FOR group had greater ADG than the CON group during wk 9 to 12 ($P = 0.04$). In contrast, significant bedding $\times$ day interactions ($P \leq 0.05$) were found for the following measures: BW (during wk 1 to 8 and wk 9 to 12), ADG and withers height (during wk 1 to 8, wk 9 to 12, and wk 1 to 12), and heart girth (during wk 1 to 12). Calves in STR group had greater ADG compared with those in RIC group for wk 9 to 12 ($P = 0.03$).

Blood Metabolites and Immunity Parameters

Several serum metabolites were measured in this experiment on the end of the wk 12 (Table 5 and Table 6). Among the metabolites, significant diet $\times$ bedding interactions were observed for glucose ($P = 0.01$) and BHB ($P = 0.03$), whereas ALB ($P = 0.10$) and GSH-Px ($P = 0.07$) tended to be significant. The FOR-RIC group had the highest serum glucose concentration, whereas the CON-RIC group had the least, although values were similar to CON-STR and FOR-STR. Calves on CON-RIC

had the highest BHB concentration and CON-STR had the least, with both forage groups being intermediate. Bedding material had an effect on the urea concentration, such that the STR group had lower urea compared with the RIC group ($P < 0.01$). The T-AOC was greater in the STR group compared with the RIC group ($P = 0.03$). Interleukin 6 ($P = 0.03$) was higher in the STR group and lower in the RIC group. Calves in the CON group had lower concentrations of haptoglobin ($P = 0.03$) and LBP ($P = 0.03$) compared with those in the FOR group. A tendency toward significance for diet $\times$ bedding interactions was observed for LPS ($P = 0.05$), which was ascribed to greater concentration of LPS in CON-RIC compared with CON-STR and FOR-STR, but similar to FOR-RIC. Neither diet nor bedding material had effect on NEFA, TP, albumin, TC, total bilirubin, T-SOD, MDA, SAA, TNF- α , or IL-10.

Rumen Fermentation

The effect of diet and bedding material on the rumen pH and TVFA is presented in Table 7 and Figure 2. A 3-way interaction ($P < 0.01$) was detected for rumen pH, whereas none was observed for the concentrations of

Table 3. Effect of diet and bedding material on nutrient intake¹

Item ³	CON		FOR		SEM	P-value ²					
	STR	RIC	STR	RIC		DIET	BED	DIET × BED	DIET × T	BED × T	DIET × BED × T
CP intake (g/d)											
wk 1 to 8	40.5	38.4	30.4	35.8	4.98	0.21	0.74	0.45	0.40	0.01	0.95
wk 9 to 12	500.2	444.4	487.0	520.1	22.4	0.25	0.47	0.08	0.01	<0.01	0.41
wk 1 to 12	193.7	173.7	182.6	193.9	10.2	0.66	0.67	0.13	0.001	<0.01	0.01
EE intake (g/d)											
wk 1 to 8	355.1	354.7	359.8	364.8	3.64	0.05	0.54	0.46	0.45	0.14	0.02
wk 9 to 12	55.3 ^{ab}	49.1 ^a	58.3 ^b	62.7 ^{bc}	2.59	<0.01	0.74	0.05	<0.01	<0.01	0.29
wk 1 to 12	255.2	252.8	259.3	264.1	2.83	0.01	0.67	0.22	0.48	0.11	0.01
Starch intake (g/d)											
wk 1 to 8	58.0	55.0	42.0	48.8	7.08	0.12	0.79	0.49	0.09	0.02	0.97
wk 9 to 12	716.7	636.6	683.0	710.0	31.9	0.54	0.41	0.10	0.02	<0.01	0.45
wk 1 to 12	277.5	248.9	255.6	269.2	14.4	0.96	0.60	0.15	0.04	<0.01	0.02
NDF intake (g/d)											
wk 1 to 8	28.0	26.6	33.1	44.6	4.40	0.01	0.26	0.15	<0.01	<0.01	0.29
wk 9 to 12	346.3 ^a	307.6 ^a	450.1 ^b	513.4 ^c	20.9	<0.01	0.56	0.02	<0.01	0.04	0.18
wk 1 to 12	134.1 ^a	120.3 ^a	172.2 ^b	200.8 ^c	9.29	<0.01	0.43	0.03	<0.01	0.19	<0.01
paNDF intake (g/d)											
wk 1 to 8	27.0	25.6	30.1	39.9	4.05	0.04	0.30	0.17	<0.01	<0.01	0.38
wk 9 to 12	334.2 ^a	296.9 ^a	416.9 ^b	470.6 ^c	18.9	<0.01	0.67	0.02	<0.01	0.02	0.19
wk 1 to 12	129.4 ^a	116.1 ^a	159.0 ^b	183.5 ^c	8.48	<0.01	0.52	0.03	<0.01	0.16	<0.01
ADF intake (g/d)											
wk 1 to 8	13.0	12.4	18.7	26.0	2.44	<0.01	0.18	0.11	<0.01	<0.01	0.12
wk 9 to 12	161.1 ^a	143.1 ^a	239.6 ^b	281.6 ^c	12.1	<0.01	0.32	0.02	<0.01	0.25	0.19
wk 1 to 12	62.4 ^a	56.0 ^a	92.3 ^b	111.2 ^c	5.28	<0.01	0.24	0.02	<0.01	0.19	<0.01

^{a-c}Within a row, means with different superscripts differ significantly ($P < 0.05$). Letter differences arise from a significant DIET × BED interaction.

¹CON = calves fed only starter concentrate; FOR = calves fed starter concentrate with oat hay; STR = straw bedding; RIC = rice husk bedding.

²DIET = dietary treatment (CON, FOR); BED = bedding treatment (STR, RIC); T = time. For all variables, data were summarized weekly; data were analyzed for wk 1 to 8, wk 9 to 12, and wk 1 to 12.

³EE = ether extract; paNDF = physically effective NDF. Feed intake was calculated excluding bedding consumption.

TVFA, acetate, propionate, and butyrate concentrations, or the acetate-to-propionate ratio (Table 7). At the end of wk 12, calves in RIC group had significantly lower rumen pH than those in STR group ($P = 0.01$). The TVFA concentration was influenced by diet, with FOR calves showing higher values than CON calves ($P = 0.05$). A 2-way interaction (bedding × time) was observed for acetate ($P = 0.05$). Although several blood parameters (e.g., TP, ALB, TC, TBill) did not differ significantly among treatments, their inclusion provides a baseline metabolic profile and supports the interpretation that major alterations in protein or lipid metabolism were not induced by the experimental treatments. The absence of differences in these parameters further underscores that the observed effects were more closely related to specific dietary and bedding interactions rather than generalized metabolic disruption.

DISCUSSION

Dietary Intake

Calves are pseudomonogastric, with a limited ability to digest fiber in the preweaning period. Thus, they depend on both milk and the starch from starter feed to meet their

nutritional needs. However, calves begin to consume either concentrates or forage, likely in an effort to supplement the energy available for growth and health. During the preweaning period, forage supplementation increased total DMI, primarily through greater forage consumption, without significantly affecting starter intake, whereas after weaning, it increased both starter intake and total DMI (Terré et al., 2013). Our results showed that milk and total solid feed and dry matter intakes were greater in calves fed concentrates with forage compared with those fed only concentrates. The observed difference in milk intake is more likely explained by divergent feeding motivation influenced by dietary composition. Calves in the FOR groups were provided with oat hay, which introduced dietary variety and sensory stimulation. This provision of forage has been shown to enhance overall feeding interest and exploratory behavior in calves (Bagheri et al., 2021), potentially creating a more robust appetite that extended to all feed offerings, including milk. Calves fed forage have been shown to have greater concentrate intake in calves fed low volumes of milk (Downey and Tucker, 2023) in some studies. However, in a study that fed their calves 6 L/d of milk, calves that were provided with chopped timothy hay had lower concentrate intake, although no differences were observed in total DMI (Hor-

Table 4. Effect of diet and bedding material on BW, ADG, feed-to-gain ratio, and structural growth¹

Item	CON		FOR		SEM	P-value ²					
	STR	RIC	STR	RIC		DIET	BED	DIET × BED	DIET × T	BED × T	DIET × BED × T
BW (kg)											
Initial (d 1)	42.4	40.4	41.1	41.0	1.07	0.73	0.33	0.35	—	—	—
Weaning	75.7	76.7	75.5	79.9	2.65	1.00	0.99	1.00	—	—	—
Final	222.5	215.5	228.4	230.9	2.72	0.01	1.00	0.96	—	—	—
wk 1 to 8	57.5	56.4	57.1	58.2	1.11	0.53	0.99	0.33	0.65	<0.01	0.38
wk 9 to 12	97.2	94.8	97.7	101.6	2.60	0.16	0.80	0.23	0.43	0.01	0.67
wk 1 to 12	67.5	66.0	67.3	69.1	1.42	0.31	0.91	0.26	0.09	0.07	0.32
wk 1 to 28	84.70	82.61	85.21	87.04	1.79	0.17	0.94	0.28	0.01	0.78	0.84
ADG (kg/d)											
wk 1 to 8	0.59	0.65	0.62	0.69	0.03	0.25	0.03	0.69	0.29	<0.01	0.04
wk 9 to 12	1.02	0.85	1.05	1.01	0.05	0.04	0.03	0.18	0.46	0.44	0.53
wk 1 to 12	0.73	0.72	0.76	0.80	0.03	0.08	0.72	0.36	0.28	<0.01	0.07
wk 1 to 28	0.78	0.76	0.81	0.84	0.03	0.05	0.85	0.35	0.35	<0.01	0.09
Withers height											
Initial (d 1)	76.5	75.2	76.3	75.6	0.71	0.91	0.17	0.70	—	—	—
wk 1 to 8	81.9	82.0	82.2	82.3	0.60	0.64	0.87	0.97	0.61	0.04	0.13
wk 9 to 12	92.2	91.0	91.8	92.1	0.82	0.62	0.63	0.37	0.59	0.05	0.90
wk 1 to 12	84.8	84.6	84.9	85.1	0.62	0.61	0.94	0.74	0.87	0.03	0.21
Heart girth											
Initial (d 1)	79.1	78.0	78.8	78.9	0.65	0.55	0.41	0.32	—	—	—
wk 1 to 8	87.9	87.2	88.4	88.6	0.62	0.14	0.67	0.44	0.31	0.29	0.83
wk 9 to 12	102.9	104.5	105.2	106.5	0.84	0.06	0.26	0.67	0.31	0.12	0.48
wk 1 to 12	92.5	92.2	93.2	93.7	0.64	0.09	0.90	0.49	0.25	0.04	0.94
FE ³											
wk 1 to 8	2.89	2.87	2.74	2.13	0.28	0.11	0.26	0.29	0.34	0.17	0.26
wk 9 to 12	2.50	2.50	2.51	2.69	0.08	0.22	0.23	0.23	0.16	0.52	0.32
wk 1 to 12	2.76	2.75	2.66	2.32	0.19	0.16	0.35	0.38	0.22	0.14	0.23

¹CON = calves fed only starter concentrate; FOR = calves fed starter concentrate with oat hay; STR = straw bedding; RIC = rice husk bedding.

²DIET = dietary treatment (CON, FOR); BED = bedding treatment (STR, RIC); T = time. For all variables, data were summarized by 2 wk; data were analyzed for wk 1 to 8, wk 9 to 12, and wk 1 to 12.

³FE = feed efficiency, calculated as total DMI divided by ADG.

vath and Miller-Cushon, 2017). Similar to this study, others have shown that calves fed high volumes of milk can have greater DMI beginning at wk 5 (Khan et al., 2011a; Castells et al., 2012). A rapid increase in total solid feed

DMI was observed in our study starting from wk 8, which may primarily be driven by 2 factors: First, systematic weaning began on d 50, with a gradual reduction in milk supply, which stimulated the intake of solid feed to meet

Table 5. Effect of diet and bedding material on blood parameters¹

Item ³	CON		FOR		SEM	P-value ²		
	STR	RIC	STR	RIC		DIET	BED	DIET × BED
Glucose (mmol/L)	6.44 ^a	5.98 ^b	6.29 ^{ab}	6.58 ^a	0.14	0.10	0.53	0.01
NEFA (mmol/L)	0.81	0.86	0.79	0.88	0.05	0.95	0.17	0.68
Urea (mmol/L)	4.23	5.00	4.16	4.77	0.20	0.46	0.001	0.69
BHB (mmol/L)	0.39	0.58	0.43	0.45	0.04	0.26	0.01	0.03
TP (g/L)	62.9	61.3	62.9	62.6	1.28	0.62	0.46	0.62
ALB (g/L)	37.0	36.3	36.9	37.7	0.47	0.16	0.88	0.10
TC (mmol/L)	1.53	1.42	1.53	1.47	0.08	0.77	0.27	0.71
TBill (μmol/L)	2.81	2.77	2.59	2.81	0.24	0.69	0.70	0.60

^{a,b}Within a row, means with different superscripts differ significantly ($P < 0.05$). Letter differences arise from a significant DIET × BED interaction.

¹CON = calves fed only starter concentrate; FOR = calves fed starter concentrate with oat hay; STR = straw bedding; RIC = rice husk bedding.

²DIET = dietary treatment (CON, FOR); BED = bedding treatment (STR, RIC). For all variables, the data were sourced from the end of wk 12.

³NEFA = nonesterified fatty acids; TP = total protein; ALB = albumin; TC = total cholesterol; TBill = total bilirubin.

Table 6. Effect of diet and bedding material on serum immunity and oxidative stress parameters¹

Item ³	CON		FOR		SEM	P-value ²		
	STR	RIC	STR	RIC		DIET	BED	DIET × BED
GSH-Px (U/mL)	166.8	155.2	151.7	163.5	6.29	0.60	0.99	0.07
T-SOD (U/mL)	139.7	133.7	137.1	134.2	4.60	0.83	0.34	0.74
MDA (nmol/mL)	3.59	3.97	4.04	4.43	0.46	0.33	0.40	0.99
T-AOC (mmol/L)	0.65	0.60	0.66	0.64	0.02	0.16	0.03	0.31
SAA (ug/mL)	237.5	225.0	249.3	235.8	11.7	0.34	0.27	0.97
Haptoglobin (ug/mL)	193.2	192.0	235.9	206.2	12.3	0.03	0.22	0.25
LBP (ug/mL)	197.3	233.8	249.2	239.8	12.6	0.03	0.29	0.07
TNF-α (pg/mL)	53.3	56.1	53.6	58.5	4.39	0.76	0.38	0.81
IL-6 (pg/mL)	64.1	73.4	64.5	75.7	4.00	0.74	0.01	0.82
IL-10 (pg/mL)	12.7	13.4	13.4	14.2	7.47	0.50	0.49	0.94
LPS (EU/mL)	0.15	0.29	0.17	0.20	0.03	0.20	0.004	0.05

¹CON = calves fed only starter concentrate; FOR = calves fed starter concentrate with oat hay; STR = straw bedding; RIC = rice husk bedding.

²DIET = dietary treatment (CON, FOR); BED = bedding treatment (STR, RIC). For all variables, the data were sourced from the end of wk 12.

³GSH-Px = glutathione peroxidase; T-SOD = total superoxide dismutase; MDA = malondialdehyde; T-AOC = total antioxidant capacity; SAA = serum amyloid A; LBP = lipoprotein-binding protein.

the energy requirements. Second, as the calves grew older and were continuously exposed to starter feed, their rumen capacity and fermentative ability improved, enabling them to consume more solid feed. Forage intake can be affected by various factors, including the type of forage and the feeding strategy (TMR or separate components; Engelking et al., 2020). The calves fed forage consumed between 17% and 21% as a percentage of the total solid, which falls in the ranges reported by Horvath and Miller-Cushon (2017). Notably, other studies have indeed reported lower forage intake proportions, sometimes below 10% of total solid intake (Hill et al., 2008; Mirzaei et al., 2015). The variation across studies can be attributed to several factors, such as the physical form and quality of the forage offered, the age at which forage is introduced, and the feeding management. It has been argued that calves fed hay with similar CP (6.4%) to our study could have stimulated concentrate intake (Horvath and Miller-Cushon, 2017). Hence, optimal utilization of milk

and solid feed, could be linked to chemical and physical characteristics of the preweaning calf diet.

Calves fed forage consumed more total solid feed DMI. Solid feed intake in calves is inhibited by the rumen capacity. However, this capacity has not been established in young calves. Neutral detergent fiber contributes to rumen fill (Dado and Allen, 1995). Our results suggest that the calves still had greater capacity to consume more forage even with greater starter intakes. Increased forage intake might indicate improved rumen environment as a result of greater intake of physically effective fiber (Castells et al., 2012; Miller-Cushon and DeVries, 2015). However, these results might not be applied across board as previous studies have indicated that forage type could limit forage intake depending on the NDF and ADF content of the fed forage (Castells et al., 2012; Miller-Cushon and DeVries, 2015). Furthermore, the type of bedding could determine whether to feed hay or not. Our results suggest that providing forage may be

Table 7. Effect of diet and bedding material on rumen pH and rumen fermentation¹

Item	CON		FOR		SEM	P-value ²					
	STR	RIC	STR	RIC		DIET	BED	DIET × BED	DIET × T	BED × T	DIET × BED × T
pH	6.39	6.20	6.43	6.21	0.08	0.71	0.01	0.85	0.02	0.50	0.01
TVFA ³ (mmol/L)	40.2	49.5	50.9	55.8	4.27	0.05	0.10	0.60	0.33	0.60	0.36
Acetate (mol/100 mol)	50.7	53.9	51.2	52.4	1.29	0.68	0.10	0.45	0.51	0.05	0.70
Propionate (mol/100 mol)	29.4	27.5	28.3	28.4	0.92	0.91	0.34	0.30	0.02	0.51	0.38
Butyrate (mol/100 mol)	14.9	13.8	15.6	14.4	0.89	0.43	0.21	0.96	0.68	0.06	0.95
Acetate/propionate ratio	1.90	2.17	2.00	2.03	0.10	0.82	0.14	0.25	0.04	0.08	0.35

¹CON = calves fed only starter concentrate; FOR = calves fed starter concentrate with oat hay; STR = straw bedding; RIC = rice husk bedding.

²DIET = dietary treatment (CON, FOR); BED = bedding treatment (STR, RIC); T = time. For all variables, the data were sourced from the end of wk 1, 5, 12, and 28.

³TVFA = total VFA.

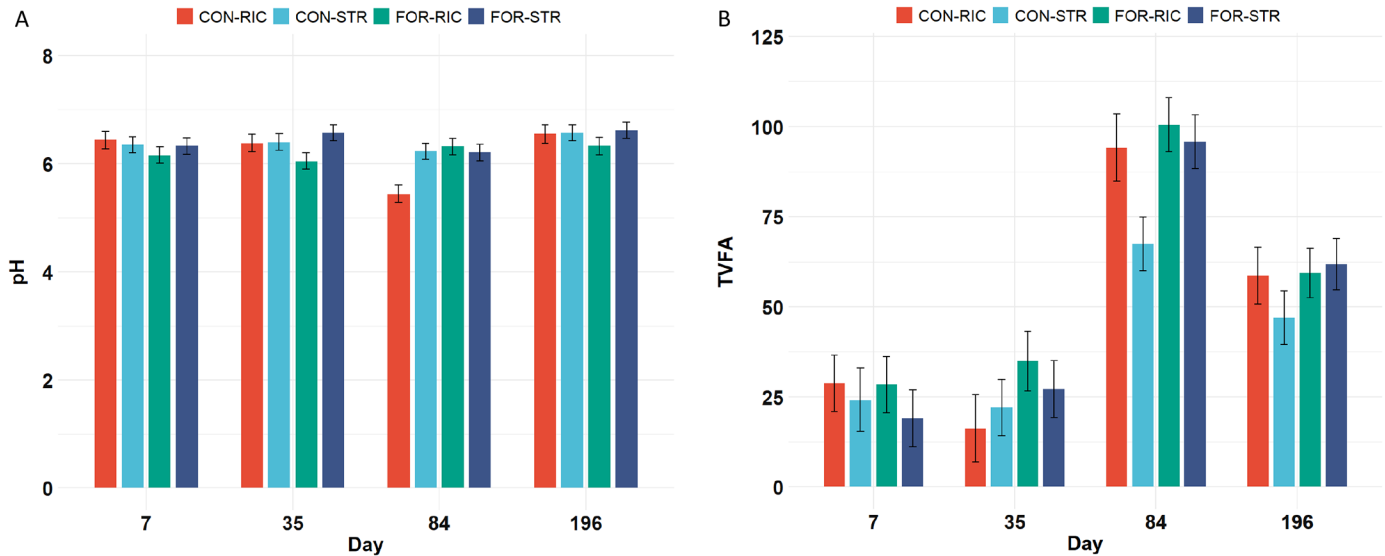


Figure 2. Effect of diet and bedding material on the rumen pH (A) and total volatile fatty acids (TVFA; B) at the end of wk 1, 5, 12 and 28. CON-RIC = calves were fed a calf starter only and bedded on rice husks; CON-STR = calves were fed a calf starter and bedded on straw; FOR-RIC = calves were fed forage in addition to a calf starter and bedded on rice husks; FOR-STR = calves were fed forage in addition to a calf starter and bedded on straw.

particularly important for increasing solid feed intake in calves bedded on rice husks.

Growth Performance and Feed Efficiency

It has been reported that calves experience anatomical development and growth of the digestive organs in response to forage feeding, a response which is reflected in greater body girth (Nemati et al., 2015). Our results showed that forage feeding improved growth rate, especially after weaning, which resulted in greater ADG in calves fed forage compared with those fed starter only in the overall period (wk 1 to 28). The observed growth rate could explain the greater BW in calves fed forage compared with those fed starter only at the wk 28. Results on effect of forage feeding on growth rates are not consistent in literature. Horvath and Miller-Cushon (2017) reported lack of differences in growth rates between calves fed forage and starter only. The effect of forage presentation method (mixed vs. separate) on calf performance remains inconsistent across studies. Some studies have found comparable intake and growth regardless of presentation method (EbnAli et al., 2016; Gasiorrek et al., 2020). However, it has been shown that feeding more than 10% hay as a mixture compared with separate components has been associated with a reduction in growth in calves (Engelking et al., 2020). This suggests that the negative impact might be dependent on the inclusion level or other specific conditions. Our results

indicate that feeding forage separately with concentrate can contribute to improved growth in the short and long term in dairy calves. Furthermore, it was reported that a greater final BW at the end of the experiment could be associated with greater development of reticulorumen in calves fed forage and high volumes of milk (Khan et al., 2011a), although growth differences were not evident in the earlier weeks. The greater withers height and heart girth at the end of the experiment in calves fed forage compared with starter only in our study corroborated with the earlier studies.

Limited studies have investigated the role of bedding materials on growth performance. A previous study reported no differences in growth parameters in calves reared on different bedding materials (Sutherland et al., 2017). Similarly, Panivivat et al. (2004) did not find differences in the growth of calves bedded on either wheat straw or rice hulls. However, Hill et al. (2011) showed that calves raised on wheat straw and housed in a nursery had better growth rates compared with those raised on sand regardless of whether they are raised in nursery or a hutches. Therefore, growth rate could be dependent on the bedding material and housing conditions. In our study, straw bedding had mixed effects on growth performance during the preweaning and postweaning periods, although the effects did not last over the long term. Most importantly, calves on rice husks bedding had greater growth rates compared with those on straw in the preweaning period, a period in which the calf immune

system is still developing. Meanwhile, both the diet and bedding material had no effect on the FE, which was consistent with previous studies.

Blood Parameters

Few studies have investigated the role of bedding materials on blood parameters. In a previous study, Sutherland et al. (2017) found no differences between treatments for various blood hematological and chemistry parameters in calves reared on either pea gravel, rubber chips, sand, or wood shavings. In our study, a few blood parameters showed differences between treatments at the end of wk 12. Glucose, TC, and total triglycerides are important indicators of body's energy status. We speculate that the differences in blood glucose could be related to the total solid feed DMI, which was shown to be numerically greater in the FOR-RIC group. Blood BHB concentration is often used as a metabolic indicator reflecting ruminal epithelial development and the initiation of ruminal fermentation in transitioning calves (Coverdale et al., 2004). However, Coverdale et al. (2004) reported that circulating BHB levels are more closely associated with age-related ruminal maturation than with the amount of solid feed consumed, and were not sensitive to differences in dietary form or forage inclusion. In the present study, calves in the CON-RIC group exhibited the highest serum BHB concentration, despite having the lowest total solid feed intake and the lowest ruminal butyrate concentration among treatments. This suggests that these calves may have had an enhanced capacity for ruminal butyrate absorption and metabolism, independent of total solid feed intake. It should be noted that other studies have reported positive correlations between BHB and starter (concentrate) intake specifically (Deelen et al., 2016; Suarez-Mena et al., 2017), but such associations are not necessarily generalizable to total solid feed intake when forage is provided. Moreover, the lack of differences in glucose between CON-RIC and FOR-STR implies that the differences in BHB were not due to decreased feed intake. Rather, the elevated BHB in CON-RIC calves most likely reflects increased circulation of BHB arising from ruminal production of ketones (Baldwin et al., 2004), which are more abundant in grain-fed calves compared with forage-fed calves. Additionally, the similar NEFA concentrations across diets suggest that serum BHB was not a result of lipolysis and subsequent hepatic ketogenesis (Terler et al., 2022).

The greater concentration in urea content is difficult to explain, as no other parameters related to protein metabolism, including TP and ALB were different when comparing bedding materials. Urea concentration in the blood is dependent on breakdown of amino acids in

the body. Thus, the greater urea concentration in calves raised on rice husks rather than straw could be ascribed to higher solid feed intake, hence protein intake in these calves.

Inflammation and oxidative stress, work together in tandem to maintain cellular homeostasis (Lei et al., 2015; Fu et al., 2024). Mammals, including calves, contain both oxidative compounds (e.g., MDA) and antioxidant systems that include T-SOD and GPX. The biological system produces reactive oxygen species during cellular activities which, if not cleared, damage cells by producing oxidative products such as MDA (Juan et al., 2021). Previous studies showed no differences in the stress related blood parameters (Sutherland et al., 2014) among calves fed either forage or concentrates. Although there were no differences in blood concentration of the oxidative stress markers across the dietary treatments, slight differences were observed in regard to bedding treatments. Although both CON-RIC and FOR-STR reported low concentration in GSH-Px, it is worth noting that CON-RIC had the lowest concentration of T-SOD and T-AOC. The lower concentration in calves raised on CON-RIC played a major role in the low total antioxidant levels in calves raised on rice husk bedding compared with straw. The T-AOC concentration reflects on the total oxidant status of the animal (Erel, 2005). Thus, the consistently low levels of T-AOC in the CON-RIC group may indicate a relatively weaker antioxidant defense system in these calves. Considering the known challenges of a high-concentrate diet, such as its potential to disrupt rumen fermentation, we speculate that this weaker antioxidant status may be related to the impact on gut health, which needs further confirmation in future studies. In contrast, calves that had exposure to forage or straw bedding might have experienced less inflammation due to increased intake of physically effective fiber in form of NDF. Hence, this is an interesting finding, in that feed type and bedding material can both have a negative compounding effect on the health of the calf. Indeed, the concentration of LPS was greater in calves on rice husk bedding compared with straw bedding, with the concentration being exceedingly greater in CON-RIC calves.

Factors such as weaning, heat stress, poor nutrition, injury, and infection can compromise immune system (Yun et al., 2014). Calves exposed to infection, tissue damage, inflammation, stress, and bacterial components are most likely to experience a surge in APP. Inflammation or presence of endotoxins in the blood have been associated with a nonspecific immune response known as the acute phase response (Werling et al., 1996). Serum amyloid A and LBP are involved directly in detoxification and removal of the systemic endotoxins (Gallay et al., 1994; Cabana et al., 1999). Studies investigating the influence

of forage feeding and bedding material on immunity parameters in calves are scarce in the literature. In a previous study, although rumen LPS in the concentrate group tended to be higher compared with the hay group, there were no differences in APP and cytokines (Takemura et al., 2019). In our study, the concentrations of SAA were similar, whereas haptoglobin and LBP concentrations were higher in calves fed forage compared with those fed concentrates.

Higher concentrations of haptoglobin have been associated with infection, tissue injury and inflammation. Liver produces haptoglobin as a result of direct stimulation by proinflammatory cytokines such as IL-6 and TNF- α (Breazile, 1988; Wassell, 2000). Based on the reported advantages of feeding forage, we expected reduced markers of inflammation like haptoglobin and SAA. On the contrary calves fed forage had increased haptoglobin concentration, but no changes in SAA in our study. Contradictory results in these 2 cytokines are not uncommon in studies, including those using mature animals (Lomborg et al., 2008). In mature cows, haptoglobin in the postcalving period was greater in cows fed a high fiber (grass hay) compared with feeding a high starch diet (Haisan et al., 2021). In an alternative study that fed timothy hay in a manner similar to the study by Haisan et al. (2021), cows fed forage had lower haptoglobin concentrations (Engelking and Oba, 2024). These 2 studies demonstrate that animals can respond differently to diets, and other factors that play a role in host response in relation to haptoglobin remain to be determined. Weaning stress could also be a contributing factor. However, long-term studies investigating the effect of weaning stress on immune parameters such as haptoglobin are scarce, and if available, no significant effects were observed (Lynch et al., 2010). Calves raised on concentrates only might have a more advanced immune defense compared with those raised on concentrates and forage. More studies are required to better understand the effects of concentrates feeding on gut health and inflammation in ruminants, as has been suggested in the previous study (Krogstad and Bradford, 2023).

Bedding material had a limited effect on the immunity parameters, as reflected in similar concentrations of APP. However, we observed greater concentrations of IL-6 in calves bedded on rice husks compared with straw. Increase in the concentrations of inflammation associated cytokines has been shown to increase with an increase in LPS challenge in steers (Carroll et al., 2009). Consistent with our results on IL-6, LPS concentration was higher in calves raised on rice husks compared with those raised on straw. The higher concentration of IL-6 in calves raised on rice husks implies that these calves had a greater need to recruit the adaptive immune sys-

tem to eliminate the endotoxin. The greater availability of physical effective fiber might partially explain these results by attenuating endotoxin release through mitigating low pH levels in calves on straw bedding. However, the lack of differences in APP responses implies that the cytokine effect was not strong enough to induce significant changes in the liver. A limitation of this study is the single blood sampling time point at the end of wk 12. This was chosen to capture the postweaning metabolic and immune steady state after adaptation to solid feed. Although it cannot delineate dynamic changes during the weaning transition, the significant differences detected in key parameters at this stage confirm that the dietary and bedding interventions had sustained effects. Future studies with serial sampling would help to elucidate the temporal dynamics of these interactions.

Rumen Fermentation

Most studies report greater rumen pH in calves supplemented with forage in the diet compared with feeding concentrate only, which is linked to a better rumen environment (Castells et al., 2013). Some studies, such as (Suárez et al., 2007), whose pH ranged between 5.1 and 5.3, did not find differences between dietary treatments in calves fed either concentrates only or concentrates with forage. Compared with Suárez et al. (2007), our pH values were significantly higher (above 6.20 for all treatments). Most importantly, significant differences were found between treatments in this study at the end of wk 12, whereby CON-RIC had the least rumen pH. The higher rumen pH observed in calves fed forage is likely attributed to greater TVFA absorption from the rumen, coupled with improved buffering capacity promoted by increased chewing and rumination activity. Other factors, such as the physical form of starter, particle size of forage, and forage intake, also regulate the rumen pH (Omidi-Mirzaei et al., 2018). However, studies investigating the role of paNDF in calves are limited (Poier et al., 2022). The “effective fiber” concept originated from adult ruminant nutrition. We applied paNDF as an exploratory tool to quantify the physical structure of fiber (NASEM, 2021). In the present study, we used paNDF as a practical, descriptive measure to compare the differences in physical fiber structure between the contrasting diets (concentrate vs. oat hay) and bedding materials (straw vs. rice husks). Under the conditions of this trial, calves fed forage consumed a greater amount of paNDF than those receiving concentrate only, which reflects the expected differences in physical diet composition. We emphasize that the relationship between paNDF and rumen development in calves remains to be established, and the findings here

should be interpreted as preliminary observations within this specific experimental context. Our results showed that calves fed forage consumed a greater amount of paNDF compared with their counterparts that were exposed to concentrates only. We speculate that low rumen pH in calves fed starter only and bedded on rice husks, compared with the other groups, could be ascribed to the former group consuming a lower amount of fiber and consequently keeping the rumen pH at a lower level. The low rumen pH was consistent with high serum LPS and low T-SOD and T-AOC concentrations in this study. The results highlighted the need to for further investigation into how much bedding material the calf can consume.

Calves fed forage generally had greater TVFA concentrations than those on starter only. It is expected that, due to greater supply of fermentable carbohydrates that are ideal for certain rumen bacteria (Omid-Mirzaei et al., 2018), calves fed starter mainly should have higher TVFA concentrations. Similar to greater TVFA concentrations, we expected calves on the forage diet to produce more acetate and less propionate due to the higher proportion of acetate-producing bacteria (Lin et al., 2018). On the contrary, diet had no effect on either parameter. However, calves bedded on rice husks had a greater proportion of acetate compared with those bedded on straw. We speculate that the observed difference arises from the distinct physical properties (e.g., hardness, shape, fiber composition) of rice husks compared with straw, which may alter calf intake or chewing behavior, thus influencing rumen fermentation profiles.

This study was designed to examine the interactive effects of bedding type and forage provision in dairy calves. Nevertheless, several methodological limitations should be acknowledged. We did not quantify actual bedding intake by the calves. Given that calves, particularly when bedded with straw, may ingest notable quantities of bedding material, it was not feasible to accurately distinguish between fecal nutrients originating from the supplied feed and those derived from consumed bedding. Future studies that incorporate edible bedding materials should include direct measurement of bedding intake to allow for accurate assessment of nutrient digestibility.

CONCLUSIONS

This study indicates that forage feeding could interact with bedding material to affect nutrient intake, growth, and health of the calves, especially in the postweaning period. Calves fed a concentrate-only diet and bedded on rice husks had the lowest feed intake, resulting in decreased nutrient intake, including the lowest levels of CP, EE, starch, NDF, and ADF. Consequently, lower blood glucose concentrations, along with an increase

in BHB, were associated with the lower average daily gain in the CON-RIC group. Additionally, these calves exhibited higher levels of LPS and IL-6, along with lower concentrations of T-AOC, suggesting increased gut inflammation. In contrast, the inclusion of forage in the diet improved both growth and health outcomes, emphasizing the necessity of providing forage for calves bedded on rice husks. Furthermore, when straw was used as bedding material, the effects of forage feeding were less pronounced.

NOTES

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Nonstandard abbreviations used: ALB = albumin; APP = acute-phase protein; CON = concentrate diet; EE = ether extract; FE = feed efficiency; FIR = ratio of forage to total solid feed intake; FOR = concentrate diet with oat hay; GSH-Px = glutathione peroxidase; LBP = lipopolysaccharide-binding protein; MDA = malondialdehyde; NEFA = nonesterified fatty acid; paNDF = physically adjusted NDF; RIC = rice husk bedding; SAA = serum amyloid A; STR = straw bedding; T-AOC = total antioxidant capacity; TBill = total bilirubin; TC = total cholesterol; TP = total protein; T-SOD = total superoxide dismutase; TVFA = total VFA.

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